🛠️ MaintainIQ

AI maintenance agent for a Tata Steel–style rolling-mill plant: predict failures from sensors, history and the production schedule, then plan maintenance. (Synthetic illustrative data.)

High-risk machines	Need action now	Expected loss (7d)
3 of 8	3	₹167.9 L

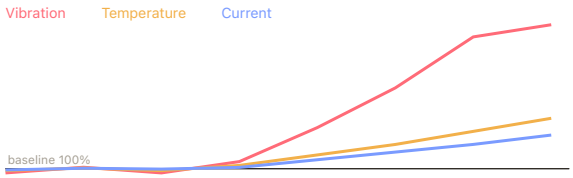
Savings from PM
₹156.5 L

Machine health – ranked by expected loss (click a row)

Machine	Line	Vibration	Overdue	Fail prob.	RUL	Exp. loss	Ac
Hot Mill Gearbox	Hot Strip Mill	6.4	133%	77%	1 d	₹82.0 L	Ac
Blast Furnace Fan	Ironmaking	3	78%	21%	30+ d	₹25.3 L	Sc wi
Descaler Compressor	Hot Strip Mill	4	111%	49%	13 d	₹24.9 L	Sc wi
Cooling Water Pump	Hot Strip Mill	2.2	130%	55%	30+ d	₹16.7 L	Ac
Slab Caster Roller	Steelmaking	2.4	46%	8%	30+ d	₹14.0 L	Mo
Crane Hoist Drive	Stockyard	3.4	161%	74%	7 d	₹4.5 L	Ac
Coil Conveyor Motor	Finishing	1.9	44%	5%	30+ d	₹0.4 L	Mo
Packaging Line					30+		

Packaging Line Drive	Dispatch	1.3	50%	6%	30+ d	₹0.1 L	Mo

Hot Mill Gearbox – Hot Strip Mill



Sensor readings as % of healthy baseline (last 8 days)

Failure pattern: **vibration**. Anomaly score 5.8/6, 3 failures in 12 months, 160 days since service (interval 120).

Recommendation: Act now – stop-and-fix within 1 days, ahead of the 5-day planned window.
Check bearings, shaft alignment and balance.

Agent

Machines at risk | Failure patterns | Maintenance plan

Ask the agent. It reasons over the live scores in the table.

What-if

Apply recommended maintenance to Hot Mill Gearbox

Planned maintenance cost ₹3 L vs. expected unplanned loss ₹82.0 L (77% failure chance).

Algorithm used by the agent

1. **Ingest** daily vibration, temperature and motor current, maintenance history (days since service, interval, failures in 12 months) and the production schedule (load, cost per downtime hour, next planned shutdown window).
2. **Baseline & anomaly**: per sensor, `z = (recent − baseline mean) ÷ baseline spread`. Anomaly score `A` = average of clipped z-scores (0–6).
3. **Overdue ratio** `O = days since service ÷ service interval`.
4. **Failure probability (7-day)** `p = 1 ÷ (1 + e^−x)` with `x = −3.2 + 0.55·A + 0.9·max(0, O−0.8) + 0.25·past failures`.
5. **Remaining useful life**: vibration trend extrapolated to the alarm limit (7.1 mm/s).
6. **Expected loss** `p × repair hours × downtime cost per hour × load`, compared to the planned-maintenance cost to get savings.
7. **Schedule**: if RUL is shorter than the next planned window, act now; otherwise bundle the work into that window to avoid peak-production downtime.

8. **Diagnose pattern** from the dominant sensor: vibration → bearing/alignment, temperature → cooling/lubrication, current → motor/electrical, none → routine service.